

The Interaction between Competition and Environmental Policy in Polluting Industries

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- Climate change has caused various policy responses, including carbon pricing.
 - Carbon pricing in place in EU, Canada, UK, China, Japan, and many others
- Growing recognition that “green innovation” and “infrastructure investments” are key for decarbonization.
 - These and emissions are equilibrium objects.
 - This has prompted policy makers to consider whether sustainability should be part of merger policy
 - Examples: EU and Australia

Big picture questions: What is the impact of competition in a polluting industry? How do competition and environmental policy interact in the pursuit of decarbonization?

- The European Commission's New Draft Merger Guidelines (published April 2026):
 - In defining the key price- and non-price parameters on which firms compete, they include prices, innovation, and sustainability (among other).
 - “It [EU Merger Control] also allows for the preservation of green innovation competition, which supports sustainability and the transition to low carbon technologies.”
- The Australian Competition and Consumer Commission (ACCC) may authorize an otherwise anticompetitive merger if it passes a “net public benefit” test
 - The ACCC explicitly recognizes environmental efficiencies such as emissions reductions, renewable energy transition, and biodiversity protection as valid public benefits.
 - *Out-of-market* efficiencies are allowed.

This Paper

We ask:

- 1 What is the equilibrium impact of competition on social welfare in the context of a polluting industry?
- 2 How does a carbon tax reshape this relationship? Does the magnitude of the tax burden matter?

Our approach:

- Oligopoly model with carbon tax
- Upstream monopolist developing an emission-abatement innovation
- The monopolist chooses how many licenses of its technology it sells

Model Setup

Market Structure

- n firms competing à la Cournot.
- Inverse demand: $P(Q) = 1 - Q$.
- Baseline carbon tax: $\tau < 1$.

Technology & Abatement

- An incremental abatement technology is available.
- Developed by a monopolist innovator choosing how many licenses to sell
 - Today: The magnitude of the innovation α is exogenous
- m adopters, each with emissions $(1 - \alpha)q_i$.
- $m - n$ non-adopters, each with emissions: q_i
- Adopters cut their tax bill (per unit produced), $C_A(q_i) = (1 - \alpha)\tau q_i$, relative to non-adopters, $C_N(q_i) = \tau q_i$.

Environmental Externality:

- Social cost of carbon: c per unit of emissions

Oligopoly Equilibrium (Given n and m)

Under the incremental innovation assumption, both types of firms produce positive quantities.

Equilibrium Quantities:

- Adopters: $q_A = \frac{1-\tau+(n-m+1)\alpha\tau}{n+1}$
- Non-Adopters: $q_N = \frac{1-\tau-m\alpha\tau}{n+1}$
- Total Industry: $Q = \frac{n(1-\tau)+m\alpha\tau}{n+1}$

Equilibrium Profits:

- Adopters: $\pi_A = (q_A)^2$
- Non-Adopters: $\pi_N = (q_N)^2$

The Innovator's Problem

The innovator sets m to maximize licensing revenue: $\Pi(m) = m\Delta(m, n)$. The incremental rent (willingness to pay) is $\Delta(m, n) = \pi_A(m, n) - \pi_N(m - 1, n)$.

The optimal licensing strategy depends strictly on the size of the carbon tax relative to a threshold, $t_\alpha = \frac{2}{2+\alpha(3n-2)}$:

Small Tax ($\tau \leq t_\alpha$)

The innovator sells to everyone: $\mathbf{m}^* = \mathbf{n}$.

Large Tax ($\tau > t_\alpha$)

The innovator restricts licenses to create scarcity: $\mathbf{m}^* = \frac{n}{4} + \frac{1-(1-\alpha)\tau}{2\alpha\tau}$.

Comparative Statics for Optimal Licenses

Focusing on the restricted licensing regime (Large Tax, $\tau > t_\alpha$):

- **Market Size (n):** $\frac{\partial m^*}{\partial n} = \frac{1}{4} > 0$.
 - For every 4 new firms entering the market, the innovator only issues 1 additional license.
- **Abatement Effectiveness (α):** $\frac{\partial m^*}{\partial \alpha} = -\frac{(1-\tau)}{2\alpha^2\tau} < 0$.
 - The larger the abatement potential, the fewer licenses sold (to maintain a higher premium per license).
- **Carbon Tax (τ):** $\frac{\partial m^*}{\partial \tau} = -\frac{1}{2\alpha\tau^2} < 0$.
 - A higher carbon tax incentivizes the innovator to sell fewer licenses.

Deconstructing Total Welfare

Total Welfare (W) captures a tension between three economic forces.

$$W = \underbrace{TS}_{\text{Total Surplus}} - \underbrace{c \times E}_{\text{Environmental Externality}}$$

- 1 **Total Surplus ($TS = CS + PS$):** $Q - Q^2/2$. The overall value generated by the market, driven entirely by total quantity Q .
- 2 **Environmental Externality ($c \times E$):** $c \times [(1 - \alpha)mq_A + (n - m)q_N]$. The aggregate social cost of production, where c is the social cost of carbon
- 3 **Composition Effect:** The technological makeup of the market (dictated by the number of licenses m), which determines whether production is clean or dirty.

The Fundamental Tension: Surplus vs. Externalities

Before considering the innovator's licensing strategy, increasing market competition (n) creates a well understood tension (Buchanan 1969):

1. Total Surplus Increases (Positive for Welfare)

As n increases, total quantity Q strictly increases. Because TS is strictly increasing in Q , more competition unambiguously increases the Total Surplus of the market.

2. Environmental Externality Increases (Negative for Welfare)

Because emissions are a byproduct of production, the total environmental damage scales with Q . Every additional unit generated by competition pumps more emissions into the atmosphere.

New Effect: The Composition Effect

Composition Effect: what kind of firm is entering the market when n increases?

This is entirely dictated by the innovator's licensing strategy (m^*):

- If the innovator grants licenses freely ($m = n$), the market composition remains **“clean”**. The emissions per unit produced stay low, easing the externality tension.
- If the innovator restricts licenses ($m < n$), the market composition turns **“dirty”**. The marginal entrant is forced to produce without abatement, causing the environmental externality to accelerate rapidly.

Welfare Synthesis: Tax Regimes Comparison

Small Tax ($\tau \leq t_\alpha$)

Market State: The tax is low enough that the innovator sells to everyone ($m = n$).

The Economic Balance as $n \uparrow$:

- **Total Surplus:** Increases as competition drives up total quantity Q .
- **Composition Effect:** Because every firm is clean, the **Environmental Externality** of the expanded production is heavily discounted by the abatement factor $(1 - \alpha)$.

Large Tax ($\tau > t_\alpha$)

Market State: The tax is high, so the innovator creates an artificial bottleneck, granting only 1 license for every 4 new firms.

The Economic Balance as $n \uparrow$:

- **Total Surplus:** Continues to rise with Q , benefiting the market.
- **Composition Effect:** The market “floods” with dirty production. The marginal entrants lack the technology, magnifying the **Environmental Externality**.

Visualizing the Inflection Point

Let's visualize these effects with specific parameters:

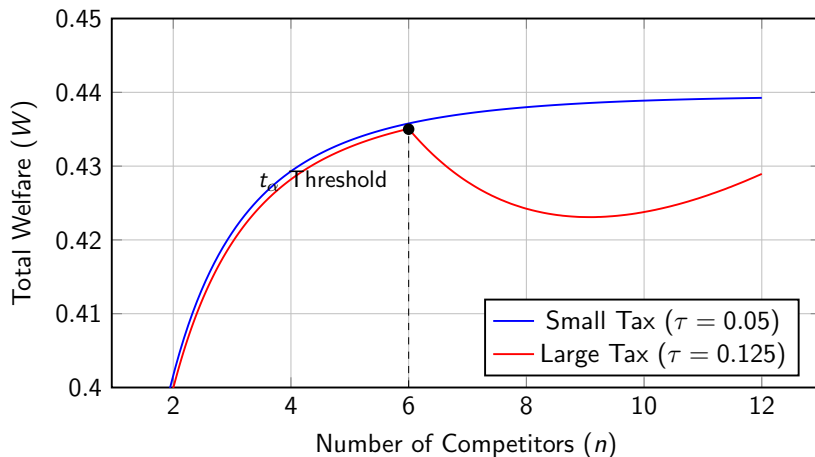
Parameters:

- Abatement Effectiveness (α): 0.875 (87.5% reduction)
- Social Cost of Carbon (c): 0.5

We compare two scenarios:

- 1 **Small Tax** ($\tau = 0.05$): The market stays in the full-adoption regime.
- 2 **Large Tax** ($\tau = 0.125$): The market crosses the licensing threshold exactly at $n = 6$.

Total Welfare vs. Market Competition (n)



Moderate Abatement: The Overproduction Problem

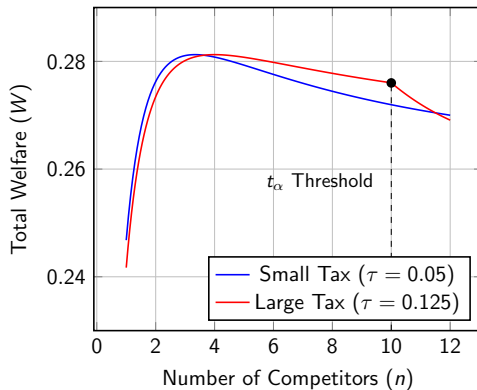
Parameters:

- Abatement: $\alpha = 0.5$ (Moderate)
- Social Cost of Carbon: $c = 0.5$

The Dynamics Swap:

Because the technology is less effective, the socially optimal total quantity drops.

As n grows, the market begins to overproduce a dirty good.



Concluding Remarks

- Competition and the carbon tax interact non-trivially in equilibrium
- The Bottleneck Effect: When the tax is too high, introducing new competitors to the market primarily introduces heavy polluters, leading to a net decrease in societal welfare.
- Next steps:
 - Endogenous α
 - Radical vs incremental innovation
 - Optimal policy design